# ORAL CANCER IMAGE CLASSIFICATION

## AND CAMERA-BASED DETECTION PROTOTYPE

*Comprehensive Technical Project Report*




Figure 1. End-to-end workflow represented by the supplied notebooks.

**Research and educational prototype — not a medical device**

# Document Scope and Evidence Basis

This report documents the code and saved outputs in the supplied notebooks. Numerical results are transcribed from notebook outputs or calculated directly from their reported confusion matrix. It distinguishes the exploratory HOG-only stage from the final HOG+LBP stage. No dataset source paper, annotation protocol, patient-level identifiers, software lock file, or licensing statement was supplied; those gaps are identified rather than filled with assumptions.

Important safety note: model labels and scores are experimental. They are not diagnoses and must not be used to rule oral cancer in or out or to delay professional evaluation.

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# Executive Summary

The project implements a two-class image classification experiment for images stored under Normal and Oral Cancer categories. It proceeds through three notebooks: (1) exploratory analysis and hyperparameter tuning of classifiers using HOG-only features, (2) final feature engineering using HOG plus LBP and a weighted soft-voting ensemble, and (3) a live-camera proof of concept that localizes a mouth region with MediaPipe landmarks and applies the saved classifier.

The final feature matrix recorded by the notebook contains 1,233 images and 8,110 features per image: 8,100 HOG values plus a normalized 10-bin LBP histogram. The final train/test split contains 986 training and 247 testing examples. The final weighted soft-voting model combines SVM, XGBoost and MLP with weights 2:1:1, in that estimator order, and labels an example Oral Cancer when the predicted class-1 score reaches 0.40.

| Final ensemble result | Recorded value |
|---|---|
| Test accuracy | 0.82996 (82.996%, approximately 83.00%) |
| ROC-AUC | 0.87711 |
| 5-fold CV macro F1 | 0.83047 ± 0.02210 |
| Oral Cancer recall at threshold 0.40 | 128/137 = 93.4% |
| Normal recall / specificity | 77/110 = 70.0% |
| Test examples | 247 (110 Normal, 137 Oral Cancer) |

The reported 83% test accuracy is a dataset-specific experimental result, not clinical validation. The final confusion matrix contains 33 Normal images predicted as Oral Cancer and 9 Oral Cancer images predicted as Normal. The camera program classifies the entire estimated mouth ROI and does not locate or segment lesions. Its results may be unreliable if camera conditions or the captured tissue differ from the training dataset.

# 1. Project Objectives and Scope

## 1.1 Objectives

- Inspect image dimensions and visualize examples from both classes.
- Standardize images to a fixed size and grayscale representation.
- Compare conventional machine-learning classifiers and search selected hyperparameters.
- Represent image structure using HOG and local texture using LBP.
- Combine three fitted estimators with weighted soft voting.
- Evaluate the ensemble using a held-out test set and stratified cross-validation.
- Export the trained ensemble and demonstrate camera-driven inference.

## 1.2 Intended scope

The implemented scope is a prototype for machine-learning experimentation and software integration. The notebooks do not establish clinical readiness, patient-level generalization, screening effectiveness, or usefulness for treatment decisions.

# 2. Three-Notebook Architecture

| Notebook | Main responsibilities | Important output |
|---|---|---|
| `analysis dataset for mouth cancer.ipynb` | Dataset inspection; resize/grayscale; HOG-only feature matrix; train/test split; GridSearchCV for several estimators; early equal-weight voting experiments. | Model comparison outputs and exploratory tuned estimators; HOG feature length 8,100. |
| `final_simulation(3).ipynb` | Repeats preparation; creates HOG+LBP features; fits XGBoost/SVM/MLP; evaluates weighted voting; saves the fitted ensemble. | `oral_cancer_voting_model.pkl`; final metrics and plots. |
| `Mouth cancer(1).ipynb` | Loads the pickle, initializes MediaPipe FaceLandmarker, opens a camera stream, crops mouth ROI, extracts matching features and overlays results. | Live windows: camera, mouth ROI and red-color mask. |

The first notebook uses HOG alone; its feature vectors contain 8,100 values. The final notebook uses HOG+LBP (8,110 values). Model results across the two stages are not a controlled head-to-head comparison because the feature representation and voting setup changed.

# 3. Dataset Inspection and Preparation

## 3.1 Expected directory structure

```
project/
├── dataset/
│   ├── normal/
│   └── Oral Cancer photos/
├── analysis dataset for mouth cancer.ipynb
├── final_simulation(3).ipynb
└── Mouth cancer(1).ipynb
```

The notebook maps `normal` to class 0 and `Oral Cancer photos` to class 1. The final feature-generation cell prints class counts `[551, 682]` in this order, totaling 1,233 samples.

## 3.2 Image-size audit

The initial dataset scan reports 1,238 readable images and 932 distinct width-height combinations. Example images have different native sizes, which motivates fixed-size preprocessing. However, the final feature matrix contains 1,233 images. The five-image discrepancy is not explained by the saved notebook outputs and should be audited by comparing folder file lists, supported extensions, image decoding failures, and generated intermediate directories.

## 3.3 Image preprocessing

1. Read class-folder image files using OpenCV.
2. Resize each accepted image to 128 × 128 pixels. This direct square resize may distort aspect ratios.
3. Convert the resized BGR image to grayscale and save it in `dataset_gray`.
4. Extract the feature representation from the grayscale image. The final feature extraction loop sorts the filenames and skips files that cannot be decoded or whose extensions are outside JPG/JPEG/PNG.

### 3.4 Partitioning and validation

The final notebook calls `train_test_split` with `test_size=0.2`, `stratify=y`, and `random_state=42`, yielding 986 training rows and 247 test rows. It uses `StratifiedKFold(n_splits=5, shuffle=True, random_state=42)` for cross-validation on the training portion. The split is image-level. Patient-level grouping is not demonstrated in the source code.

# 4. Feature Engineering

### 4.1 Histogram of Oriented Gradients

HOG summarizes local gradient orientations and is used to encode edges and local shape/structure. The OpenCV descriptor parameters in both feature pipelines are: window size 128×128, block size 16×16, block stride 8×8, cell size 8×8, and 9 orientation bins. OpenCV reports a descriptor length of 8,100 values per image.

| HOG parameter | Value |
|---|---|
| Window size | 128 × 128 |
| Block size | 16 × 16 |
| Block stride | 8 × 8 |
| Cell size | 8 × 8 |
| Orientation bins | 9 |
| Descriptor length | 8,100 |

### 4.2 Local Binary Pattern

The final notebook also computes a uniform Local Binary Pattern (LBP) image with P=8 neighbors and radius R=1. A histogram is computed with 10 bins and normalized by its sum (the code adds a small epsilon in the training pipeline). The final feature extraction code therefore contributes 10 texture histogram values.

### 4.3 Combined representation

| Feature block | Length | Role (general description) |
|---|---|---|
| HOG | 8,100 | Local gradient/edge structure |
| Normalized LBP histogram | 10 | Compact local texture statistics |
| Concatenated feature vector | 8,110 | Input to final classifiers |

The training and camera inference code reproduce the same HOG+LBP ordering and test the expected feature count of 8,110. Mixing the HOG-only exploratory models with the final camera pipeline would cause a feature-dimension mismatch and is not supported.

# 5. Exploratory Model Selection (Notebook 1)

The first notebook uses HOG-only features. It splits the same 1,233 feature rows into 986 training and 247 test examples, then evaluates several algorithms. GridSearchCV uses five-fold stratified cross-validation; depending on the model, selection scoring is accuracy or macro F1.

## 5.1 Decision Tree

The search considers Gini/entropy, maximum depth 2/5/10, minimum split 2/5, and minimum leaf 1/2/5 (36 combinations). The saved output reports best parameters: Gini, max_depth=5, min_samples_leaf=1, min_samples_split=5; best CV accuracy 0.7272 and test accuracy 0.6923.

## 5.2 Random Forest

The search considers 100/200/300 trees, Gini/entropy, depth 2/5/10, minimum split 2/5 and minimum leaf 1/2/5 (108 combinations), with `class_weight="balanced"`. Best recorded settings are entropy, 300 estimators, depth 10, min_samples_split=5 and min_samples_leaf=2. Best CV accuracy is 0.8286, test accuracy 0.8259 and ROC-AUC 0.8699.

## 5.3 XGBoost

The initial grid searches 8 combinations over n_estimators 100/200, max_depth 3/5 and learning_rate 0.05/0.1, with subsample and colsample_bytree 0.8, reg_alpha 0.1, reg_lambda 5 and `eval_metric="logloss"`. Recorded best CV accuracy is approximately 0.8297; test accuracy is 0.8219 and ROC-AUC is 0.8630.

## 5.4 Support Vector Machine

The exploratory SVM is wrapped with StandardScaler and uses an RBF kernel, balanced class weights and probability estimates. The displayed best parameters are C=1, gamma=`scale`, kernel=`rbf`; the selected cross-validation score is 0.8314 (the grid is configured to refit macro F1). Recorded test accuracy is 0.8340 and ROC-AUC is 0.8554.

## 5.5 Logistic Regression

A StandardScaler + LogisticRegression pipeline uses the SAGA solver and balanced class weights. The grid compares L1/L2 and elastic-net variants, C values, and selected convergence settings. Best recorded parameters include C=0.1, l1_ratio=0.75, max_iter=1000, elastic-net penalty and tolerance 0.001. Best CV macro F1 is 0.8123, test accuracy 0.7854 and ROC-AUC 0.8301.

## 5.6 MLP

The MLP grid searches three hidden-layer arrangements, with ReLU, alpha=0.001, early stopping and up to 5,000 iterations. The selected architecture is `(256, 256, 128, 64, 32, 16, 8, 4, 2)`; recorded best CV macro F1 is 0.8298. The class report corresponds to about 80.16% test accuracy and ROC-AUC 0.8239.

| Exploratory HOG-only experiment | Test accuracy | CV selection result | ROC-AUC |
|---|---|---|---|
| Decision Tree | 69.23% | 0.7272 CV accuracy | Not emphasized in summary |
| Random Forest | 82.59% | 0.8286 CV accuracy | 0.8699 |
| XGBoost | 82.19% | 0.8297 CV accuracy | 0.8630 |
| SVM | 83.40% | 0.8314 CV macro F1 | 0.8554 |
| Logistic Regression | 78.54% | 0.8123 CV macro F1 | 0.8301 |
| MLP | ≈80.16% | 0.8298 CV macro F1 | 0.8239 |
| SVM + XGBoost soft voting | 81.78% | No CV score reported for ensemble here | 0.8674 |
| SVM + XGBoost + MLP soft voting | 82.59% | No CV score reported for | 0.8626 |

| Exploratory HOG-only experiment | Test accuracy | CV selection result | ROC-AUC |
|---|---|---|---|
| | | ensemble here | |

The first notebook explores equal-weight voting variants on tuned HOG-only estimators. Its early model called `rf` in one estimator tuple actually contains the tuned MLP estimator; the report treats that as a naming inconsistency in the code, not as an additional Random Forest component.

# 6. Final Model and Voting Ensemble (Notebook 2)

## 6.1 Final base estimators

| Estimator | Recorded final settings |
|---|---|
| XGBoost | n_estimators=200;<br>max_depth=3;<br>learning_rate=0.1;<br>reg_alpha=0.1;<br>reg_lambda=5;<br>subsample=0.8;<br>colsample_bytree=0.8;<br>random_state=42;<br>eval_metric=logloss. |
| SVM | SVC with RBF kernel;<br>C=1; gamma=scale;<br>class_weight=balanced;<br>probability=True;<br>random_state=42;<br>wrapped in StandardScaler pipeline. |
| MLP | hidden_layer_sizes=(256,128,64,32,16,8,4,2);<br>ReLU; Adam;<br>alpha=0.001;<br>max_iter=5000;<br>early_stopping=True;<br>n_iter_no_change=100;<br>random_state=42;<br>wrapped in StandardScaler pipeline. |

## 6.2 Soft voting and weights

The final VotingClassifier has estimator order `[SVM, XGBoost, MLP]`, `voting="soft"`, `weights=[2, 1, 1]`, and `n_jobs=-1`. In soft voting, estimator probability outputs are combined with the specified weights. SVM therefore receives twice the voting weight of each other base estimator. The classifier is fit on the 986-row training split and saved with joblib as `oral_cancer_voting_model.pkl`.

## 6.3 Decision threshold

The final notebook explicitly predicts class 1 when `predict_proba(X_Test)[:, 1] >= 0.40`. Its comments note that ordinary `VotingClassifier.predict` would correspond to its default decision behavior instead. The source does not document a separate validation protocol for selecting 0.40. The threshold is therefore an experimental hyperparameter and should be fixed using validation data before final unbiased evaluation in future work.

### 6.4 Cross-validation nuance

The recorded cross-validation score is calculated with `cross_val_score(..., scoring="f1_macro")` on the VotingClassifier. This scoring call uses the estimator's ordinary `predict` method, whereas the final held-out test metrics are computed from the custom 0.40 threshold. Consequently, the CV macro-F1 summary and thresholded test classification report use different decision rules and should not be treated as a like-for-like threshold-specific comparison.

# 7. Evaluation Protocol and Metrics

### 7.1 Metrics in the notebook

| Metric | Interpretation |
|---|---|
| Accuracy | Fraction of all test images classified correctly. |
| Precision for Oral Cancer | Of images labelled Oral Cancer by the model, fraction that are Oral Cancer in the labelled test set. |
| Recall / sensitivity for Oral Cancer | Of true Oral Cancer images, fraction identified by the model. |
| Specificity | Of true Normal images, fraction identified as Normal; equals Normal-class recall when class 1 is positive. |
| F1-score | Harmonic mean of precision and recall. |
| Macro F1 | Unweighted average of per-class F1 scores. |
| ROC-AUC | Threshold-independent ranking summary based on the test score ordering; not a calibration measure. |
| Confusion matrix | Counts of correct and incorrect predictions per actual/predicted class. |

### 7.2 Confusion-matrix convention

For matrices shown here, class order is `[Normal (0), Oral Cancer (1)]`. Thus, rows correspond to actual classes and columns to predicted classes. When referring to sensitivity and specificity, Oral Cancer is treated as the positive class.

# 8. Final Results

The following figures and table transcribe the saved output in the final notebook. Values are tied to the recorded split and current implementation; rerunning with changed files or library versions may produce different results.

| Model | Test accuracy | Macro precision | Macro recall | Macro F1 | ROC-AUC | CV macro F1 (mean ± SD) |
|---|---|---|---|---|---|---|
| XGBoost | 0.8097 | ≈0.82 | ≈0.80 | ≈0.80 | 0.8878 | 0.8103 ± 0.0229 |
| SVM | 0.8219 | ≈0.83 | ≈0.81 | ≈0.82 | 0.8647 | 0.8339 ± 0.0107 |
| MLP | 0.7854 | ≈0.78 | ≈0.78 | ≈0.78 | 0.8648 | 0.8152 ± 0.0314 |
| Weighted voting | 0.8300 | ≈0.85 | ≈0.82 | ≈0.82 | 0.8771 | 0.8305 ± 0.0221 |

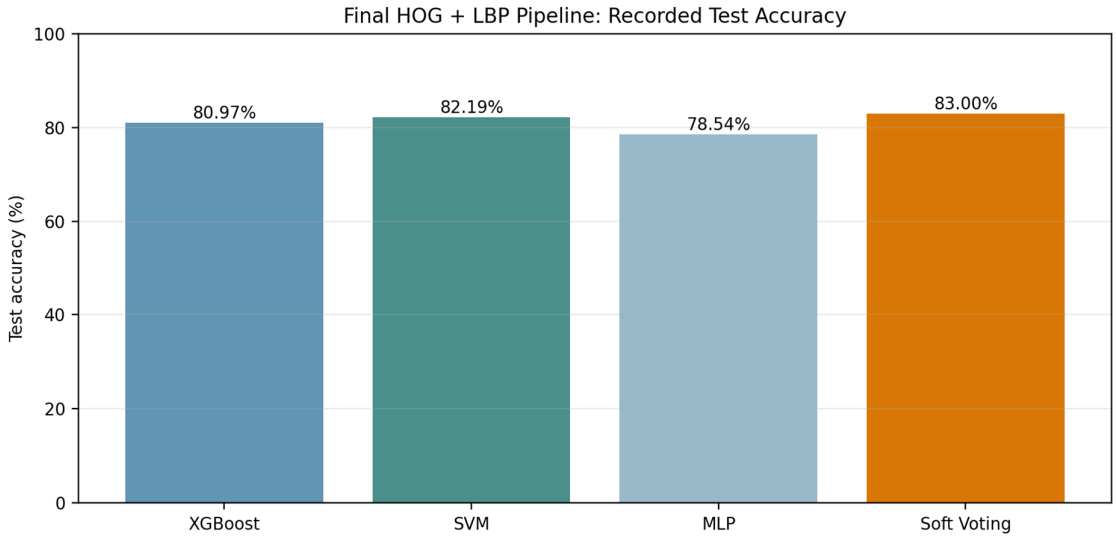



Figure 2. Recorded test accuracy for the final HOG+LBP pipeline. Accuracy figures are transcribed from notebook outputs/chart values.

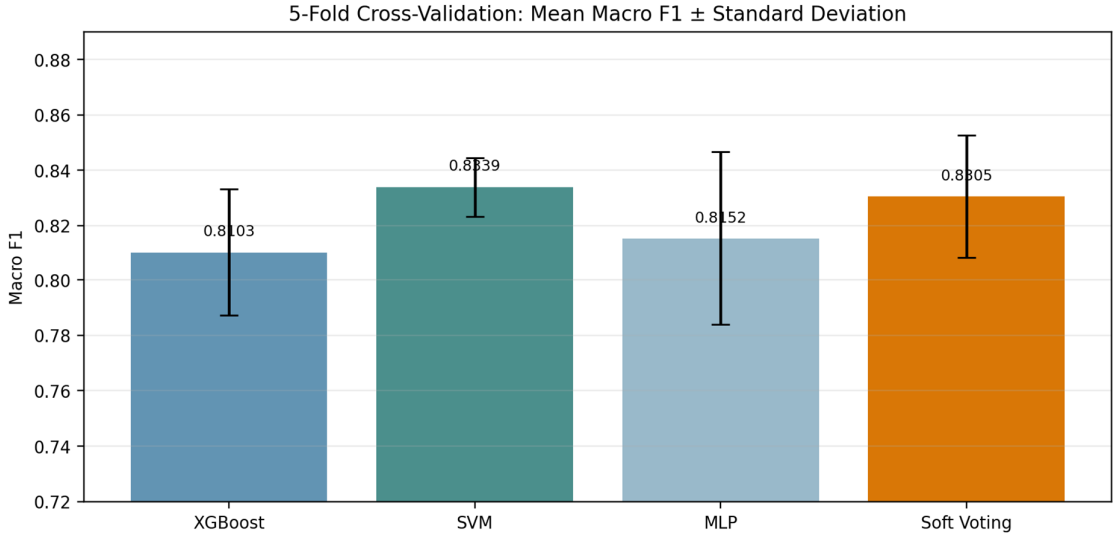



Figure 3. Recorded five-fold macro-F1 mean and standard deviation on training folds. The final notebook reports the individual fold scores.

## 8.1 Final voting classification report

| Class / metric | Precision | Recall | F1-score | Support |
|---|---|---|---|---|
| Normal (0) | 0.90 | 0.70 | 0.79 | 110 |
| Oral Cancer (1) | 0.80 | 0.93 | 0.86 | 137 |
| Accuracy | — | — | 0.83 | 247 |
| Macro average | 0.85 | 0.82 | 0.82 | 247 |
| Weighted average | 0.84 | 0.83 | 0.83 | 247 |

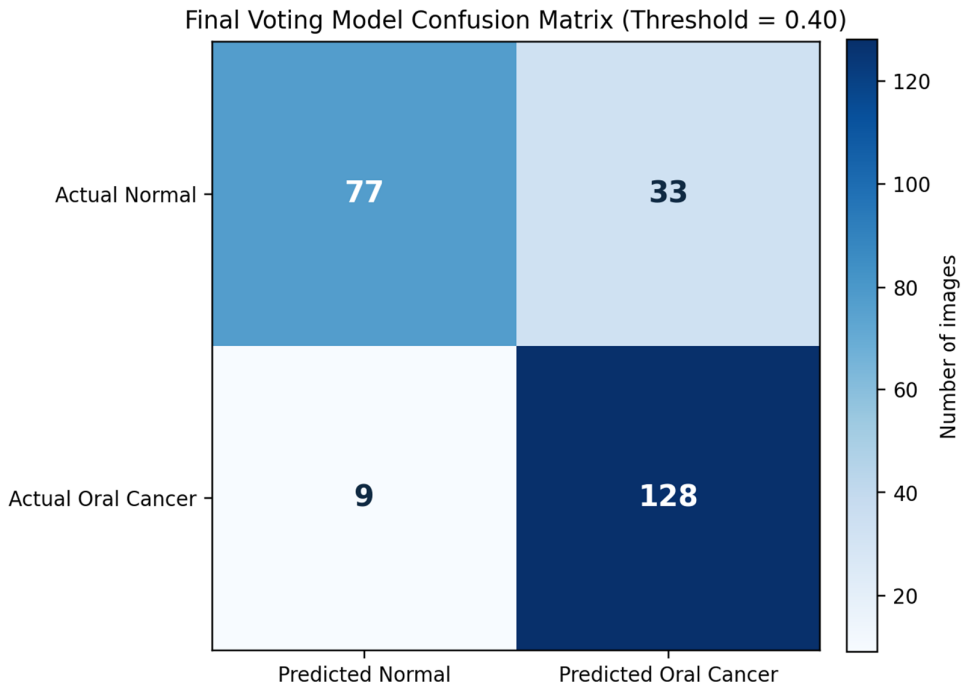



Figure 4. Final weighted voting confusion matrix, with threshold 0.40. Rows are actual classes; columns are predictions.

## 8.2 Interpretation of final errors

| Actual class | Predicted Normal | Predicted Oral Cancer | Total |
|---|---|---|---|
| Normal | 77 (correct) | 33 (false positives) | 110 |
| Oral Cancer | 9 (false negatives) | 128 (correct) | 137 |
| Total | 86 | 161 | 247 |

Derived from the recorded matrix: sensitivity = 128/137 = 93.4%; specificity = 77/110 = 70.0%; precision for Oral Cancer = 128/(128+33) = 79.5%; negative predictive value = 77/(77+9) = 89.5%; false-positive rate = 33/110 = 30.0%; false-negative rate = 9/137 = 6.6%. These quantities describe this fixed test split only; predictive values will vary with real-world prevalence and dataset composition.

The thresholded ensemble has higher Oral Cancer recall than Normal recall. This comes with 33 false-positive Normal examples in the test set. For a medical screening context, these trade-offs require explicit clinical goals, an independent validation cohort and expert review; no such clinical operating-point validation is shown in the notebooks.

## 8.3 Selected-model comparison charts and interpretation

This subsection expands the model comparison using figures from the final HOG+LBP notebook and metrics calculated from its printed test confusion matrices. Unless indicated otherwise, each figure describes the same held-out test partition of 247 images (110 Normal, 137 Oral Cancer). These results are from one recorded split and are not an estimate of clinical effectiveness.

### 8.3.1 Test accuracy comparison

Figure 5 compares the fraction of all held-out images classified correctly. The weighted voting ensemble is highest at approximately 83.00%, followed by SVM at 82.19%, XGBoost at 80.97%, and MLP at 78.54%. The ensemble's advantage over SVM is small (about 0.81 percentage points), so the figure should not be interpreted as proof that voting is statistically superior. A larger independent test set and confidence intervals would be required to assess whether this difference is meaningful.

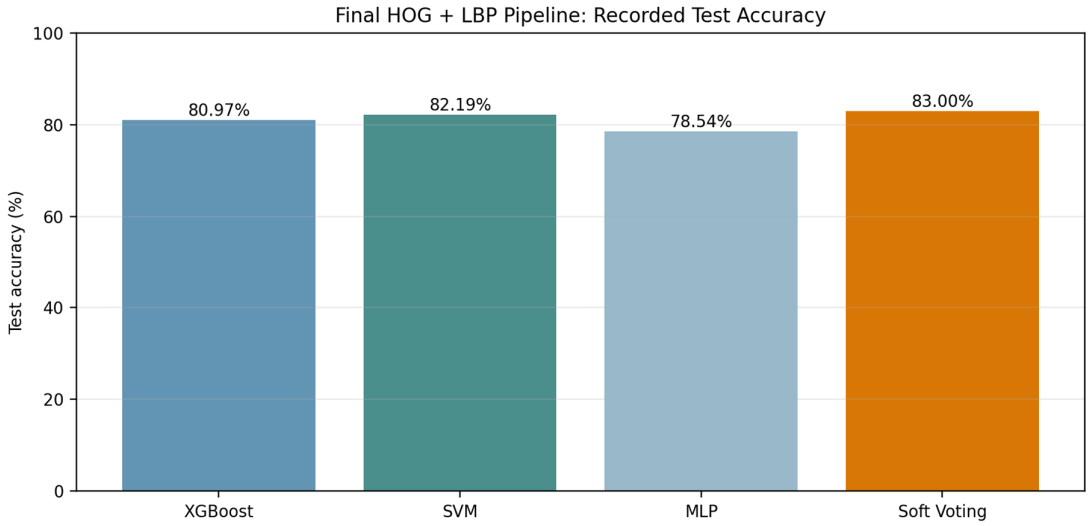



Figure 5. Test accuracy of the three final base classifiers and the weighted soft-voting ensemble.

### 8.3.2 Precision, recall and F1 for the Oral Cancer class

Figure 6 plots positive-class metrics calculated from the recorded confusion matrices rather than using rounded values typed into the notebook's comparison chart. Precision answers: "Among images predicted as Oral Cancer, what fraction were labelled Oral Cancer?" Recall/sensitivity answers: "Among Oral Cancer-labelled images, what fraction did the model identify?" F1 balances precision and recall. The ensemble has the highest recall (128/137 = 93.4%) and highest positive-class F1 (approximately 0.855), but precision is about 79.5%. In this test set, 33 Normal images are false positives. MLP has slightly higher precision (about 81.3%) but lower recall (79.6%); this illustrates the different error trade-offs rather than a universally best model.

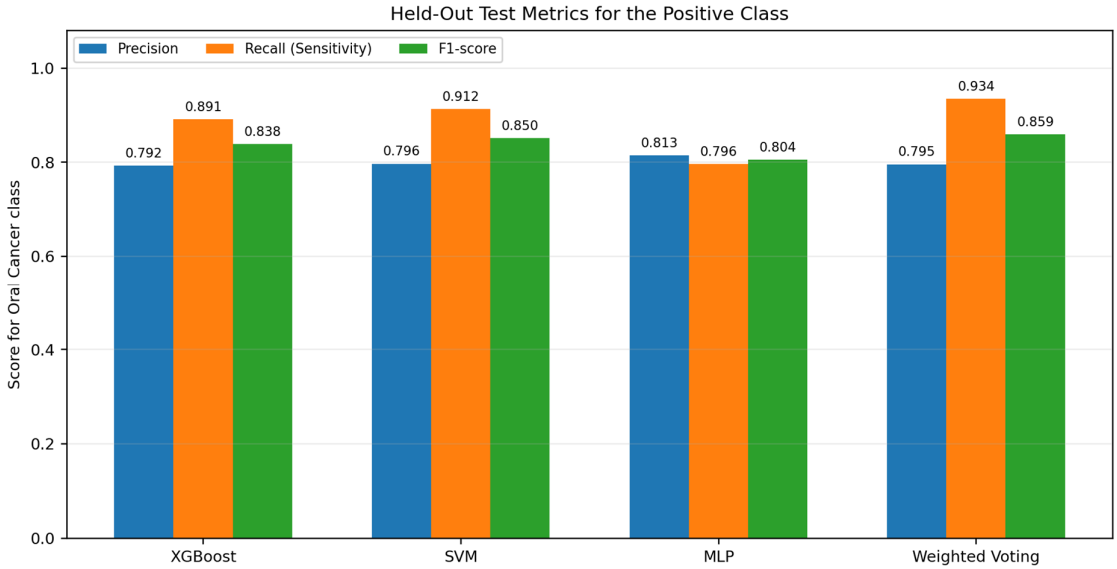



Figure 6. Positive-class precision, recall/sensitivity and F1 calculated from each model's reported test confusion matrix.

### 8.3.3 Cross-validation stability

Figure 7 shows the mean macro F1 and standard deviation across five stratified folds on the training partition. SVM has the highest recorded mean (0.8339) and the smallest standard deviation (0.0107), suggesting the most consistent fold-level macro-F1 among these four candidates in this particular cross-validation run. Voting has mean 0.8305 with standard deviation 0.0221. MLP varies most (standard deviation 0.0314). These fold scores use each estimator's ordinary prediction rule; the final ensemble's held-out test predictions instead use a manually coded probability threshold of 0.40, so CV macro F1 is not a threshold-matched validation of that operating point.

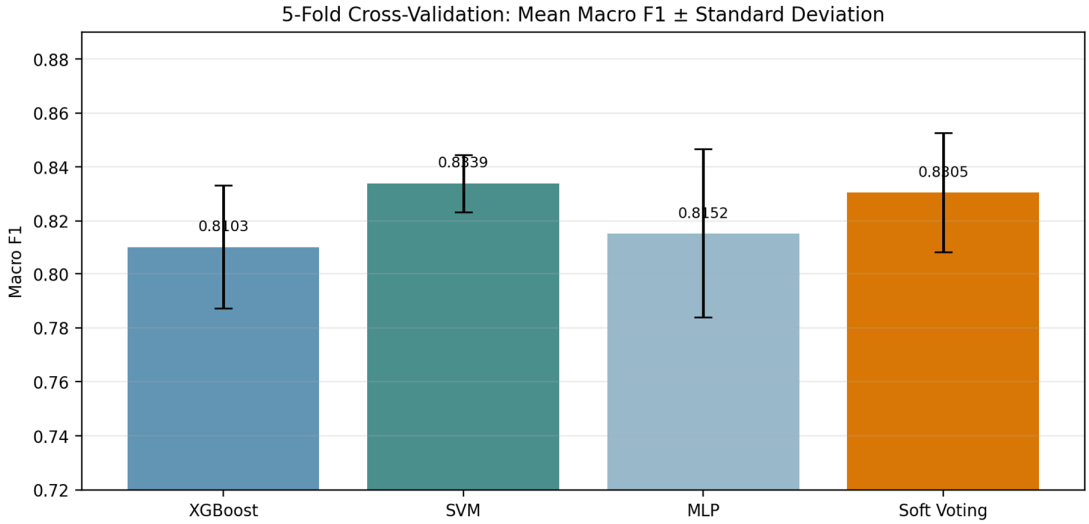



Figure 7. Five-fold training cross-validation macro F1, shown as mean ± standard deviation.

### 8.3.4 Confusion matrices and error profile

Figure 8 places the four model confusion matrices side by side. Rows are actual labels and columns are predictions; the diagonal cells are correct predictions and off-diagonal cells are errors. For the Oral Cancer class, false negatives are images labelled Oral Cancer but predicted Normal. The voting model produces 9 false negatives, fewer than XGBoost (15), SVM (12) and MLP (28) on this split. However, it produces 33 false positives, comparable to

XGBoost and SVM (32 each) and higher than MLP (25). At the chosen threshold, the ensemble therefore favours catching more positive-labelled images at the cost of flagging more Normal images.

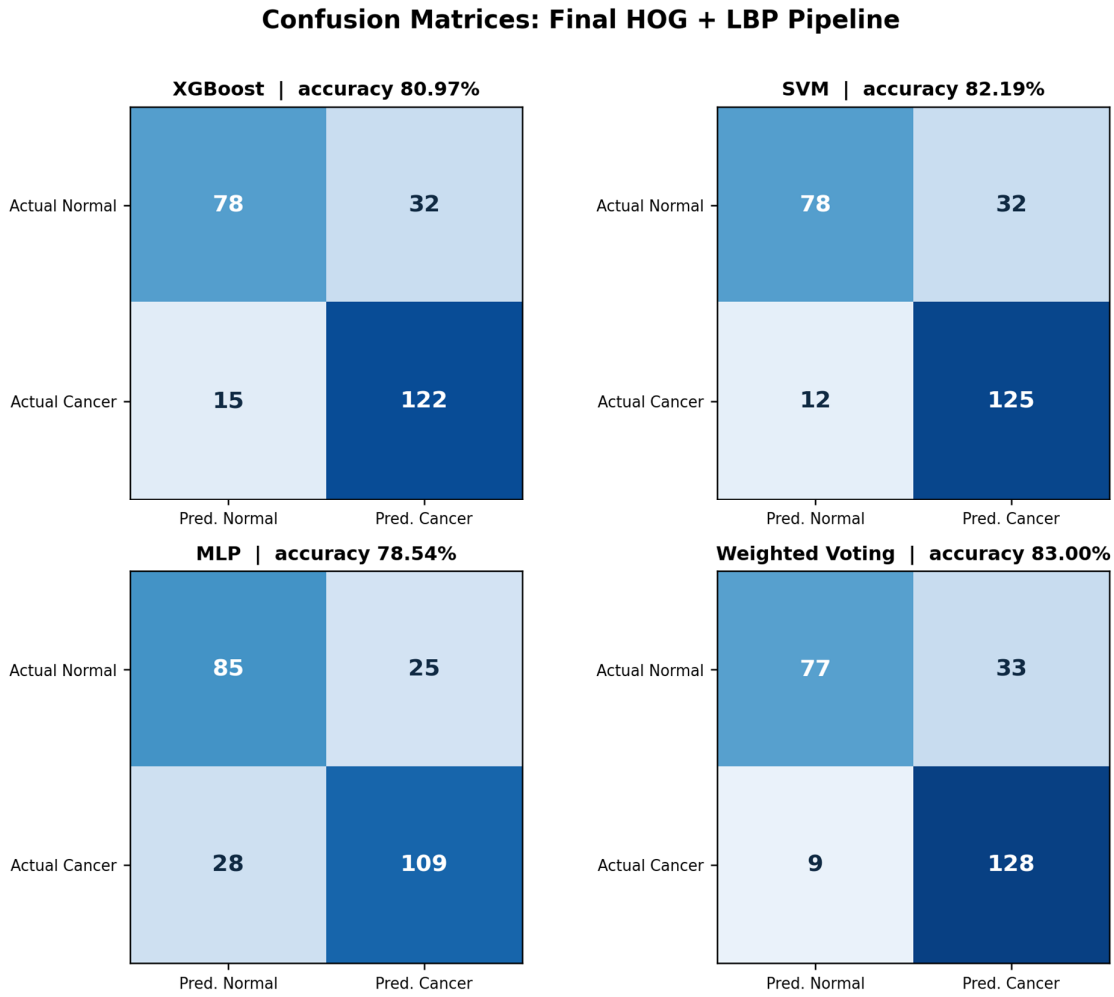



Figure 8. Four test-set confusion matrices, with class order Normal then Oral Cancer.

### 8.3.5 ROC curves and ROC-AUC

Figure 9 reproduces the ROC-curve plots stored in the final notebook. A ROC curve plots true-positive rate against false-positive rate as the decision threshold varies. A larger area under the curve (AUC) indicates stronger ranking separation across thresholds; it does not specify the final operating threshold and does not measure probability calibration. XGBoost has the largest reported ROC-AUC (0.8878), followed by voting (0.8771), MLP (0.8648) and SVM (0.8647). Thus, voting leads in test accuracy and positive-class recall at the selected 0.40 threshold, but not in ROC-AUC. This is not contradictory: these metrics evaluate different properties of the score ranking and the chosen hard-decision threshold.

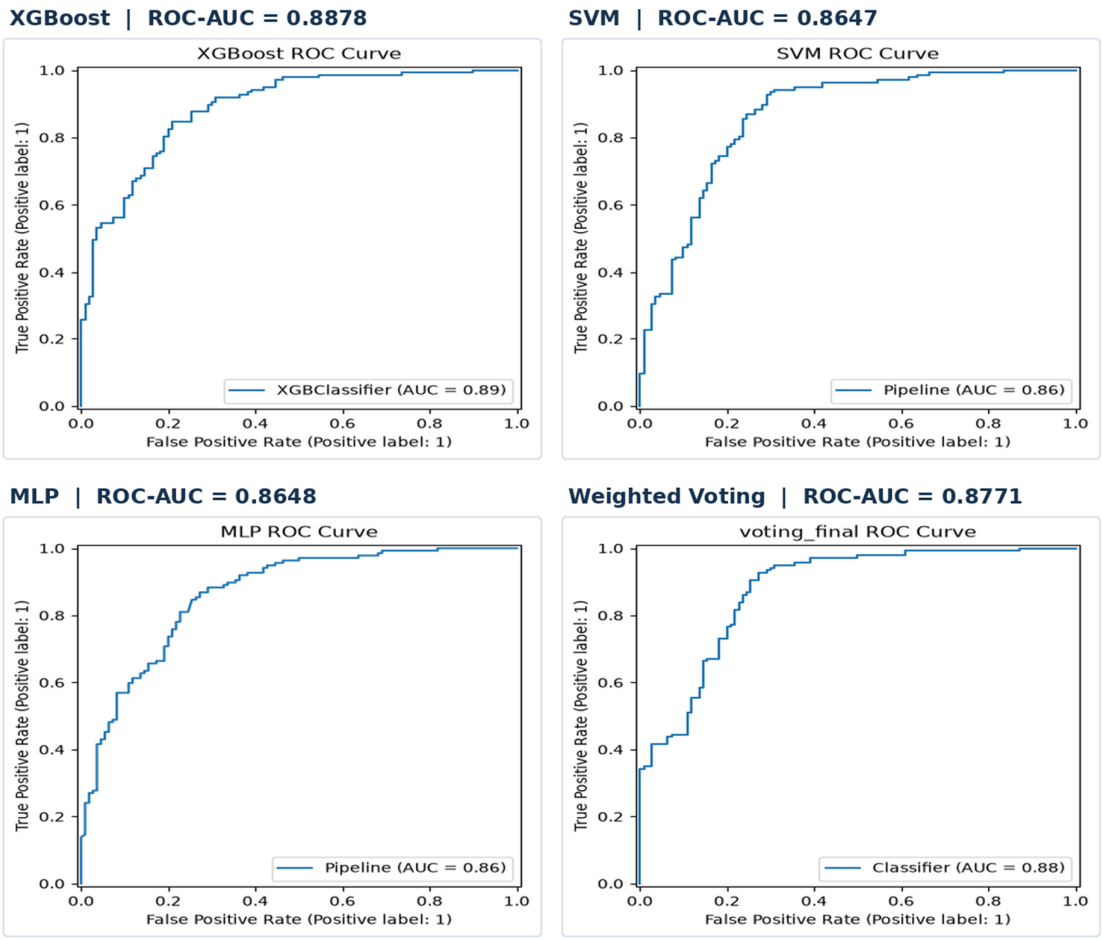



Figure 9. ROC curves reproduced from the final notebook's saved output images. AUC values are those printed by the notebook.

### 8.3.6 Precision–Recall curves

Figure 10 reproduces the final notebook's Precision–Recall plots. These show precision against recall as the score threshold changes and are especially informative when the positive class is the focus. Moving toward higher recall can reduce precision because more borderline examples are labelled positive. The curves help visualize this trade-off beyond the single threshold used in the confusion matrices. The saved outputs do not print average precision values for every model, so this report does not assign numerical PR-AUC values based on visual inspection.

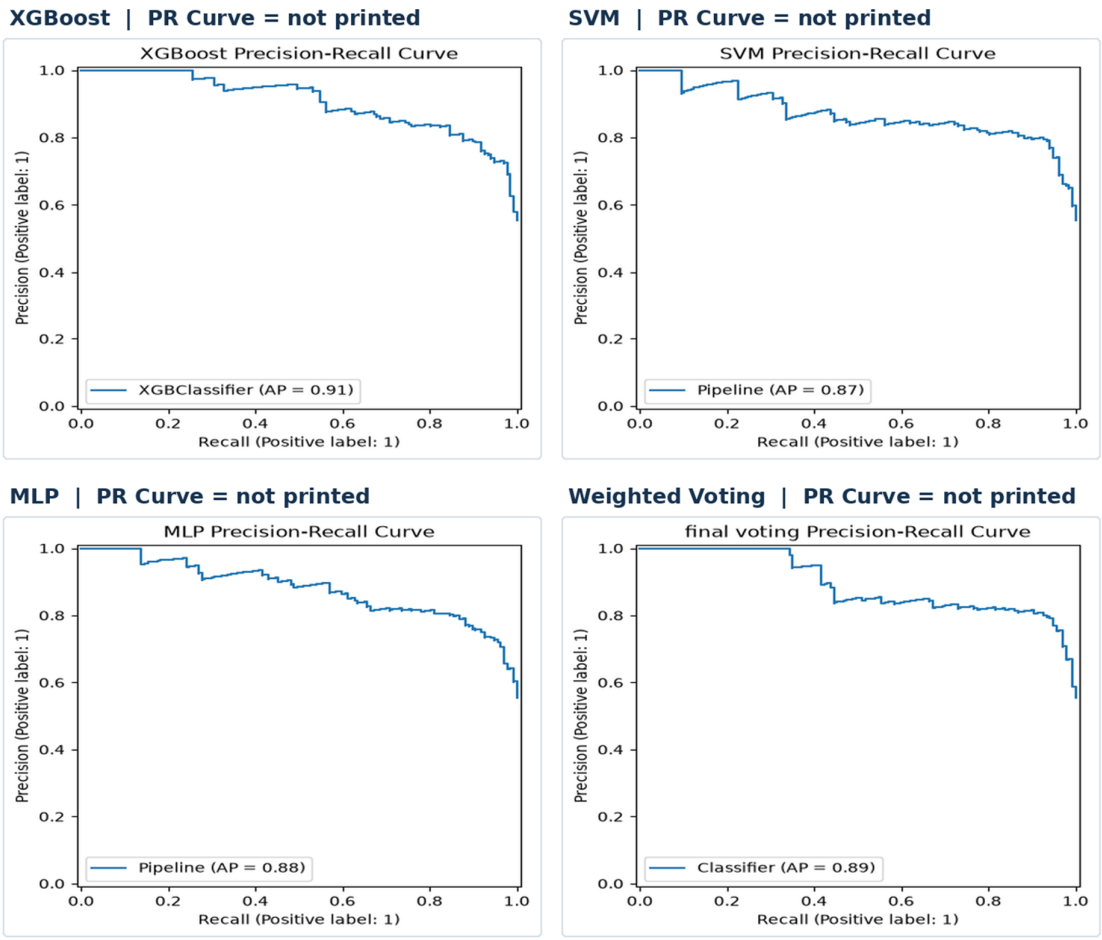



Figure 10. Precision–Recall curves reproduced from the saved final notebook outputs; no unreported area values have been inferred.

### 8.3.7 XGBoost feature-importance plot

The final notebook also plots the 20 feature indices with the largest XGBoost importance scores. This view indicates which numeric dimensions the fitted tree ensemble used most strongly within this dataset. The feature indices are not direct anatomical regions or lesion attributes, and feature importance alone cannot show causal evidence or clinical interpretability. Because the representation concatenates HOG and LBP, an index needs to be mapped to the extraction layout before it can be attributed to a feature family.

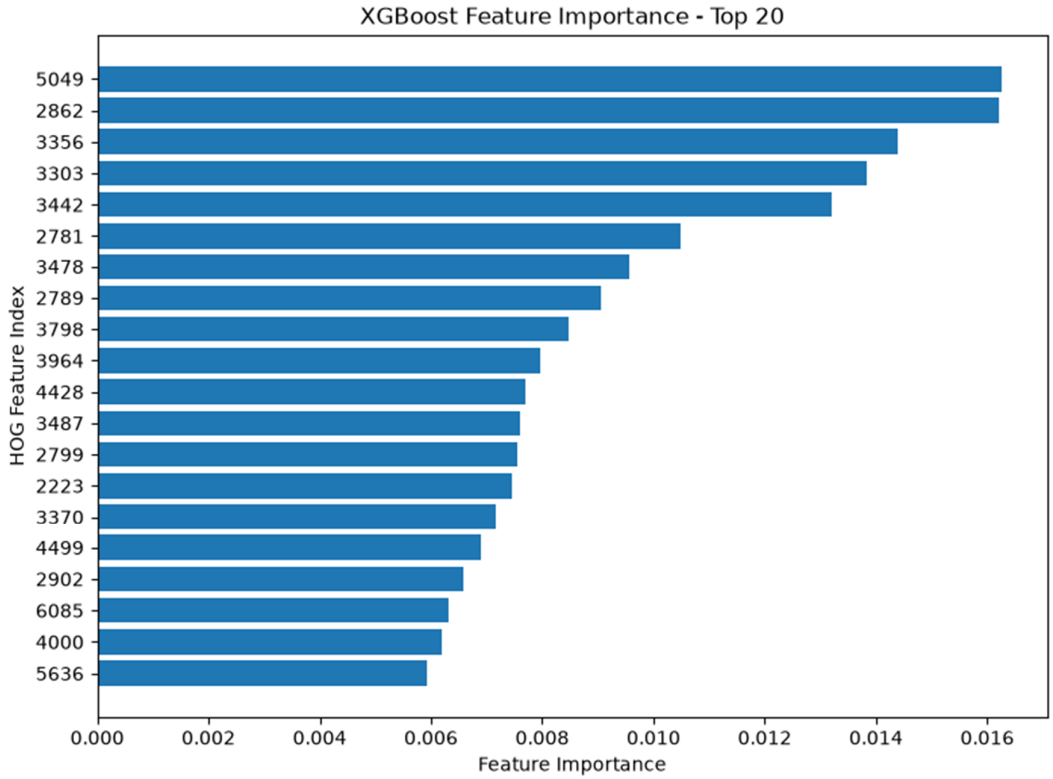



Figure 11. Top 20 XGBoost feature indices as rendered by the final notebook.

### 8.3.8 ROC-AUC comparison

Figure 12 summarizes the ROC-AUC numbers printed by the final notebook. XGBoost performs best on this threshold-independent ranking metric, while the weighted voting model is second. The ensemble is still useful because it offers a different operating-point trade-off and the highest test accuracy in these outputs. Selecting a deployed model should be based on a predefined validation criterion and error costs, not one metric or a single test split.

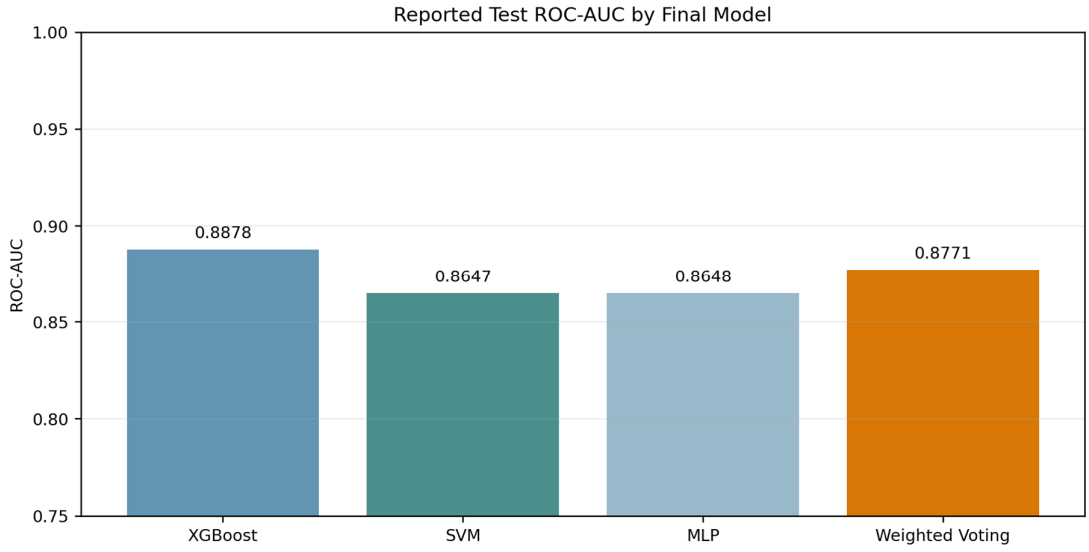



Figure 12. Reported ROC-AUC comparison for the final HOG+LBP models.

#### 8.3.9 Overall interpretation

Taken together, the saved figures show that no model is best on every criterion. The soft-voting model gives the highest thresholded accuracy and catches the largest fraction of Oral Cancer-labelled test images, while XGBoost has the highest reported ROC-AUC and SVM has the strongest mean cross-validation macro F1 with the smallest fold-to-fold standard deviation. The model choice therefore depends on the intended use and the relative cost of false negatives versus false positives. Because this is a prototype in a medical domain, the threshold and the model must be evaluated on independent, patient-separated, clinically representative data before any screening claim can be justified.

# 9. Camera-Based Inference (Notebook 3)

## 9.1 Model loading and feature consistency

The camera notebook loads `oral_cancer_voting_model.pkl` using joblib. It builds the same 128×128 HOG descriptor and computes a normalized 10-bin LBP histogram, then concatenates the features. It verifies that the combined feature count is 8,110 and, where model metadata is available, checks that the model expects the same number of features. It also verifies that class IDs 0 and 1 are present.

## 9.2 MediaPipe mouth ROI

MediaPipe FaceLandmarker uses VIDEO mode, a supplied `face_landmarker.task` model asset, and `num_faces=1`. Twenty-two mouth/lip landmark IDs are mapped from normalized landmark coordinates to pixel coordinates. The bounding rectangle is expanded by 10% of mouth width on either horizontal side, 15% above and 20% below, then clipped to frame boundaries. That cropped patch is used as the classifier input.

## 9.3 Camera stream and display

The stream is opened using `cv2.VideoCapture(CAMERA_URL, cv2.CAP_FFMPEG)`. The code checks that the stream opens and that a usable frame arrives within up to 20 attempts with 0.2-second waits. Each frame is converted from BGR to RGB for MediaPipe, timestamped monotonically, processed for face landmarks, then the mouth crop is passed through `predict_oral_cancer`. The display shows Normal and Oral Cancer scores as percentages and the thresholded model result. Press Q to stop; the `finally` block releases the camera, closes the landmarker and destroys OpenCV windows.

## 9.4 Red-color mask

The code also converts the mouth ROI to HSV and builds two red-hue masks, combines them, and applies morphological opening and closing with a 5×5 kernel. This mask appears in a separate window as a visualization only. It is not used as a feature by the machine-learning classifier and does not prove that a lesion is present.

## 9.5 Camera limitations

- The configured URL is a hard-coded example (`http://192.168.1.101:4747/video`) and must be replaced with the user's active stream address.
- A general webcam crop may not match the type, magnification, focus, lighting or tissue visibility of training images. The source code does not demonstrate camera-domain validation.

- The code classifies the complete mouth ROI rather than locating a suspicious lesion.
- If no face is detected, the system displays a message rather than generating a classification. A face landmark is not proof that the mouth region is diagnostically adequate.

# 10. Reproducibility and Execution Guide

## 10.1 Dependencies

Imports across the notebooks include NumPy, pandas, OpenCV, Matplotlib, scikit-learn, scikit-image, XGBoost, joblib and MediaPipe. Jupyter/ IPython is required to run the notebooks interactively. Use compatible versions, especially for MediaPipe Tasks APIs, scikit-learn serialization and XGBoost.

```
python -m pip install --upgrade pip
python -m pip install numpy pandas opencv-python matplotlib scikit-learn scikit-image xgboost joblib
mediapipe notebook ipykernel
```

The notebooks do not supply a pinned requirements/lock file. The recorded outputs include scikit-learn deprecation warnings for `SVC(probability=True)` and LogisticRegression parameter usage in the saved runtime. Record exact versions and test code compatibility before upgrading packages. Loading a joblib pickle created with substantially different library versions may be unsafe or fail; use the same environment where possible.

## 10.2 Recommended execution order

5. Place the two class folders under `dataset/` with the exact names and capitalization expected by the notebook.
6. Run `analysis dataset for mouth cancer.ipynb` to reproduce the exploratory HOG-only model search and ensemble experiments.
7. Run `final_simulation(3).ipynb` from a clean kernel, top to bottom. Verify the 1,233 feature rows, feature dimension 8,110, test split size 247, model metrics and successful creation of `oral_cancer_voting_model.pkl`.
8. Place the saved pickle and the compatible `face_landmarker.task` file in the camera notebook's working directory.
9. Update `CAMERA_URL`, then run `Mouth cancer(1).ipynb`. Confirm the stream URL and permissions before expecting live inference.

## 10.3 Key camera settings

```
CAMERA_URL = "http://192.168.1.101:4747/video"
MODEL_PATH = "oral_cancer_voting_model.pkl"
LANDMARKER_PATH = "face_landmarker.task"
IMAGE_SIZE = (128, 128)
LBP_POINTS = 8
LBP_RADIUS = 1
PROBABILITY_THRESHOLD = 0.40
```

## 10.4 Reproducibility checklist

- Record the dataset version and file list/checksums.
- Resolve the 1,238-versus-1,233 image-count mismatch.

- Audit duplicate and near-duplicate images and split by patient if patient identifiers are available.
- Record Python and library versions (for example, `pip freeze`).
- Run from a clean kernel and verify random seeds and split settings.
- Save the trained pickle together with a model metadata file documenting feature version, label mapping, threshold and dependency versions.
- Evaluate threshold choices on validation data and reserve the test set for the final locked evaluation.
- Store camera URL configuration separately from the code rather than committing private LAN addresses.

# 11. Limitations, Risks and Threats to Validity

- Dataset provenance and labels: the notebooks do not cite a dataset paper, specify the label adjudication protocol, or describe the source population. The reliability and scope of the labels therefore cannot be independently assessed from the supplied files.
- Image leakage: the saved split is stratified by image, not by patient. If multiple pictures of the same person/lesion or near duplicates are present, train/test overlap could inflate performance.
- Sample and domain generalization: the final test set has 247 images and its composition may not reflect intended real-world use. No independent external test set is reported.
- Preprocessing: resizing to a square can distort shapes; grayscale removes color cues; no image augmentation or image-quality rejection step is shown in the final notebook.
- Feature limitations: HOG and LBP are hand-designed features. Their success depends on image acquisition and how closely training imagery matches camera ROI crops.
- Threshold and calibration: threshold 0.40 is coded but the method for choosing it is not reported. `predict_proba` scores are not shown to be calibrated estimates of clinical risk.
- Metric protocol: cross-validation scores use default prediction behavior, while final test predictions use the 0.40 threshold. This difference should be corrected in a future evaluation design.
- Class error consequences: the final test confusion matrix has 9 false negatives and 33 false positives. Neither error count estimates clinical outcomes outside this test sample.
- Inference mismatch: MediaPipe defines the ROI from lips, but the model’s usefulness depends on what tissue is actually visible. The notebook does not demonstrate localization of a lesion or clinical image acquisition quality.
- No clinical validation: no prospective study, clinical workflow assessment, ethics review, or regulatory analysis is provided in these notebooks.

# 12. Recommendations for Future Work

## 12.1 Dataset and evaluation

- Document dataset origin, inclusion criteria, label definitions, expert review, permissions and licensing.
- Audit corrupt files, skipped extensions, duplicate images and possible near-duplicates; explain the five-image discrepancy.

- Use group-aware train/validation/test splits at patient level where patient identity is known.
- Use an independent external test set collected under different acquisition conditions.
- Report confusion matrices, sensitivity, specificity, predictive values, ROC-AUC, precision-recall AUC, confidence intervals and calibration.
- Choose the operating threshold using validation data only; lock it before final test reporting.
- Use consistent threshold rules for cross-validation and test-set metrics, or report both clearly as separate analyses.

### 12.2 Model and deployment

- Compare the handcrafted HOG+LBP representation with a properly validated image-based deep-learning baseline if dataset size and permissions permit.
- Avoid square distortion where possible by testing aspect-preserving resize and padding.
- Add image-quality checks for blur, exposure, framing, mouth opening and visible tissue.
- Evaluate different cameras, lighting settings, pose and distances; document failure rates as well as classification metrics.
- Add confidence/uncertainty messaging and a prominent disclaimer; do not provide reassurance or diagnostic language.
- Package the preprocessing function with the model so the training/inference transformation cannot drift.
- Create automated tests for class mapping, feature dimension, expected files, camera initialization and graceful cleanup.

## 13. Conclusion

The notebooks demonstrate a coherent experimental software pipeline: image inspection, preprocessing, handcrafted feature extraction, conventional classifier comparison, weighted soft-voting, model serialization and camera integration. The final HOG+LBP ensemble records 82.996% test accuracy and ROC-AUC 0.8771 on the specified image-level split. It identifies 128 of 137 Oral Cancer-labelled test images at the coded 0.40 threshold, while also mislabelling 33 of 110 Normal images as Oral Cancer. These results may justify further controlled research, but do not establish clinical performance or readiness for use with patients.

The most important next steps are to clarify dataset provenance and the image-count discrepancy, prevent patient-level leakage, define a validation-based threshold protocol, perform external evaluation, and test the compatibility between the training images and camera-derived mouth crops.

# Appendix A. Configuration and Hyperparameters

| Item | Final notebook value |
|---|---|
| Image classes | 0 = Normal; 1 = Oral Cancer |
| Training samples | 986 |
| Testing samples | 247 |
| Training feature dimension | 8,110 |
| Image preprocessing | 128×128, grayscale |
| HOG descriptor | 128×128 window; 16×16 block; 8×8 stride; 8×8 cell; 9 bins |
| LBP | P=8, R=1, method=uniform; normalized 10-bin histogram |
| Split | 80/20 stratified; random_state=42 |
| Cross-validation | 5-fold stratified; shuffle=True; random_state=42 |
| XGBoost | 200 trees; depth 3; lr 0.1; alpha 0.1; lambda 5; subsample/colsample 0.8 |
| SVM | RBF; C=1; gamma=scale; balanced class weights; probability=True; StandardScaler |
| MLP | (256,128,64,32,16,8,4,2); ReLU; Adam; alpha=.001; max_iter=5000; early stopping; n_iter_no_change=100; StandardScaler |
| Voting | Soft; estimator order SVM, XGBoost, MLP; weights [2,1,1] |
| Threshold | Class 1 when Oral Cancer score ≥ 0.40 |
| Model artifact | oral_cancer_voting_model.pkl |
| Camera asset | face_landmarker.task |
| Exit key | Q |

# Appendix B. Confusion Matrices and Derived Counts

All matrices use class order [Normal, Oral Cancer] and follow scikit-learn's row=actual, column=predicted convention.

| Final model | Confusion matrix | Interpretation (TN, FP, FN, TP) | Test accuracy |
|---|---|---|---|
| XGBoost | [[78, 32], [15, 122]] | 78, 32, 15, 122 | 80.97% |
| SVM | [[78, 32], [12, 125]] | 78, 32, 12, 125 | 82.19% |
| MLP | [[85, 25], [28, 109]] | 85, 25, 28, 109 | 78.54% |
| Weighted voting, 0.40 | [[77, 33], [9, 128]] | 77, 33, 9, 128 | 83.00% |

For binary classification treating Oral Cancer as positive: TN = Normal correctly predicted as Normal; FP = Normal predicted as Oral Cancer; FN = Oral Cancer predicted as Normal; TP = Oral Cancer correctly predicted as Oral Cancer. Values in this appendix are computed from saved notebook confusion-matrix output.

# Appendix C. Troubleshooting

| Problem | Checks / remedy |
|---|---|
| Cannot load pickle | Run final notebook, check `MODEL_PATH` and current working directory. |
| Missing MediaPipe asset | Provide a compatible `face_landmarker.task` file and verify `LANDMARKER_PATH`. |
| Camera does not open | Check URL, host IP, port, phone application, same network/firewall and FFmpeg support. |
| Camera opens but gives no frames | Confirm the stream endpoint is correct and active; inspect network reachability. |
| Feature count mismatch | Use final HOG+LBP model and identical feature settings; do not mix the HOG-only model. |
| No face detected | Improve lighting, face camera, adjust distance/framing; no classification is produced until a face ROI exists. |
| Different outputs after rerun | Check dataset files, processed folders, versions, random seed, clean kernel and pickle timestamp. |
| Library warnings/errors | Record package versions; update deprecated parameters carefully and rerun evaluation after changes. |

# Appendix D. Glossary

| Term | Definition in this project |
|---|---|
| HOG | Histogram of Oriented Gradients, a handcrafted descriptor of local image gradients. |
| LBP | Local Binary Pattern, a local texture descriptor represented here as a normalized histogram. |
| SVM / SVC | Support Vector Machine / scikit-learn support vector classifier. |
| XGBoost | Gradient-boosted decision-tree implementation used as a base estimator. |
| MLP | Multi-Layer Perceptron, a feed-forward neural network classifier. |
| Soft voting | Ensembling method combining class-probability outputs with optional estimator weights. |
| ROI | Region of interest; the image crop extracted around the mouth. |
| ROC-AUC | Area under the receiver operating characteristic curve. |
| Macro F1 | Average F1 across classes, giving each class equal weight. |
| Sensitivity | Recall for the positive Oral Cancer class. |
| Specificity | Recall for the Normal class when Oral Cancer is the positive class. |
| Calibration | Agreement between predicted probabilities and observed outcome frequencies; not evaluated in the supplied notebooks. |